

Blackbox optimization model

- ▶ Model described in [Thomas et al., 2023]
- ▶ $\mathbf{x} = (x_1, y_1, x_2, y_2, \dots, x_{N_T}, y_{N_T})$ = locations of the N_T turbines
- ▶ N_D and N_S : Number of discretized values for wind direction and speed
- ▶ Maximization of the Annual Energy Production (AEP):

$$f(\mathbf{x}) = -AEP(\mathbf{x}) = - \left[\frac{\text{hours}}{\text{day}} \right] \left[\frac{\text{days}}{\text{year}} \right] \sum_{i=1}^{N_D} \sum_{j=1}^{N_S} f_{ij} \sum_{k=1}^{N_T} P_{ij}(x_k, y_k)$$

- ▶ $P_{ij}(x_k, y_k)$: power for the turbine k with a wind direction i and speed j (computed with PyWake [Pedersen et al., 2023])
- ▶ f_{ij} the probability of a given wind speed and wind direction combination

Constraint 1: Placement of the turbines in the boundary zone and in the constructible zone

- ▶ **Boundary zone:** This corresponds to the bounds of $\mathbf{x}$ (set $\mathcal{X}$)
- ▶ **Constructible zone:**
 - ▶ **Binary version:** (x_i, y_i) must be admissible for all $i = 1, 2, \dots, N_T$: this is easily checked by a routine before launching PyWake (**A priori** binary constraint). Infeasible designs are not sent to PyWake and a blackbox evaluation is not counted.
 - ▶ **Continuous version:**

$$\sum_{k=1}^{N_T} d_k \leq 0$$

where d_k is the distance between turbine k and the closest constructible zone. This is an **A priori**, **relaxable**, and **quantifiable** constraint. The infeasible points are sent to PyWake and a blackbox evaluation is counted.

- ▶ **Variant based on a projection:** Turbines are projected to the nearest constructible zone

Constraint 2: Minimal space between two wind turbines

$$S_{ij} \geq 2d \text{ for all } i, j = 1, 2, \dots, N_T \text{ and } i \neq j$$

- ▶ S_{ij} : Spacing between turbines i and j
- ▶ d : Diameter of a wind turbine
- ▶ This is an **A priori** and **quantifiable** constraint
- ▶ Two variants: **relaxable** or **unrelaxable**
- ▶ Current version: $N_T(N_T - 1)/2$ constraints
- ▶ Alternate version: One aggregated constraint

Illustration of a solution

